THE CARBON CYCLE

Chapter 13 • Topic 1 | Answers to Questions 1 – 7

IGCSE / Grade 8 Science • Pointwise Model Answers

Question 1

Name the gas that is a reactant in photosynthesis.

• Carbon dioxide (CO_2)

Note: Water is also a reactant, but the question asks for the gas.

Question 2

Describe how the carbon atoms are present in photosynthesis. Which reactant contains carbon and which product contains carbon?

• The reactant that contains carbon is **carbon dioxide (CO_2)**.

• The product that contains carbon is **glucose ($C_6H_{12}O_6$)**.

• During photosynthesis, carbon atoms from carbon dioxide are used to make glucose (sugar).

Word equation: carbon dioxide + water → glucose + oxygen

Question 3

Explain why photosynthesis is an important factor in climate change.

• Photosynthesis takes carbon dioxide out of the atmosphere.

• Carbon dioxide is a greenhouse gas that contributes to global warming / climate change.

• By removing CO_2 from the air, photosynthesis helps reduce the amount of greenhouse gas in the atmosphere.

• Plants therefore act as a natural **sink** for carbon dioxide.

Key idea: Less CO_2 in the atmosphere → less trapping of heat → helps slow climate change.

Question 4

Suggest why strategies to reduce the effects of climate change usually include plans to grow more plants.

• Growing more plants increases the rate of photosynthesis.

• More photosynthesis means more carbon dioxide is removed from the atmosphere.

• Plants act as a larger carbon sink.

• This helps reduce the concentration of the greenhouse gas CO_2 and therefore helps reduce global warming / climate change.

Question 5

Describe how animals are sources of carbon dioxide both when they are alive, and after they have died.

When animals are alive:

• Animals carry out respiration (aerobic respiration).

• In respiration, glucose + oxygen → carbon dioxide + water.

• Carbon dioxide is released into the atmosphere.

After animals have died:

• Dead animals are decomposed by bacteria and fungi (decomposers).

• During decomposition, carbon dioxide is released into the atmosphere.

Summary: Alive → respiration releases CO_2; Dead → decomposition releases CO_2. Both make animals a source of carbon dioxide.

Question 6

During the daytime, plants produce both carbon dioxide and oxygen. During the night, plants produce carbon dioxide but not oxygen. Explain why this occurs.

During the daytime:

• Plants carry out both photosynthesis and respiration.

• Photosynthesis produces oxygen (and uses CO_2).

• Respiration produces carbon dioxide (and uses oxygen).
• Therefore both gases are produced.

During the night:

• There is no sunlight, so photosynthesis stops.
• Respiration continues (plants need energy all the time).
• Respiration produces carbon dioxide but no oxygen.
• Therefore only carbon dioxide is produced at night.

Key point: Photosynthesis needs light → only happens in the day. Respiration does not need light → happens day and night.

Question 7

'Although plants produce carbon dioxide through respiration, they are still a very useful sink of carbon dioxide because they take in twice as much carbon dioxide as they produce.' What does this statement tell you about the reactions of photosynthesis and respiration in plants?

• The rate of photosynthesis is greater than the rate of respiration in plants.
• Plants take in more carbon dioxide during photosynthesis than they release during respiration.
• Overall, plants remove more CO2 from the atmosphere than they put back.
• This is why plants act as a net sink for carbon dioxide.

In simple words: Photosynthesis removes CO2 faster than respiration adds it back → plants help reduce atmospheric CO2.

Quick Revision – Key Ideas for the Carbon Cycle

• Photosynthesis: CO2 + H2O → glucose + O2 (takes place in chloroplasts, needs light)
• Respiration: glucose + O2 → CO2 + H2O (takes place in mitochondria, day and night)
• Photosynthesis = sink of carbon dioxide
• Respiration & decomposition = sources of carbon dioxide
• Plants are still a useful sink because photosynthesis rate > respiration rate

Model answers based on the Grade 8 / Stage 9 textbook (Chapter 13 – The Carbon Cycle). Written in clear IGCSE-style pointwise format for easy learning and exam use.